

TinyOS: A Historical Review of its Motivations and Successes

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Abstract—

I. INTRODUCTION

Around the time that TinyOS was created, sensor networks were an emerging area of interest, with many fields of study working on implementing a wide range of sensor networks for their purposes. As such, creating a way for sensors to communicate with each other in a network was a great topic of interest. Because of this demand, TinyOS was created in collaboration with the University of California, Berkeley (UC Berkeley), Intel Research, and Crossbow Technology in 2000 to facilitate the development and innovation of embedded systems, specifically those consisting of many sensor nodes. Initially, it was created to deal with these problems: limited resources on motes, reactive concurrency, need for flexibility, and power demands.

Around the time when TinyOS was created, motes did not have the resources that the devices today. Instead, those devices were extremely limited in resources, and thus, each power consumption, memory, and other resources needed to be efficiently spent.

Additionally, the growing need to create a way for the system to react to incoming data needed to be addressed. In the areas that embedded systems were initially implemented, which included environmental monitoring and for military use, those systems needed a way to respond to the stimulus in a way that efficiently spent resources and responded within timing constraints.

Moreover, flexibility was an important aspect of an operating system for this task. Specifically, the operating system needed to support fine-grain modularity so that sensors connected to the network could be adapted to suit the network in which it was deployed.

Lastly, power demands were another driving factor for the creation of TinyOS. Motes needed a way to run on low-power, since at the time of creation, motes were limited in resources. The use of computational resources directly resulted in energy consumption. As such, to ensure long-term operation, an operating system that consumed less power was necessary.

II. IMPLEMENTATION

The first iteration of TinyOS was written in nesC, which is a version of C that was created specifically for TinyOS to

optimize based on the memory limits of the sensor networks. Additionally, TinyOS is non-blocking, which means that it has one call stack. Because it only has one stack, this required any input or output operation that lasted more than a set amount of time to be done elsewhere and given a callback once the operation was completed. This allowed data to move within the stack efficiently without having to wait until lengthy operations were completed. In light of these unique factors, the authors made sure to deal with the four issues listed, which, again, included: limited resources on motes, reactive concurrency, and the need for flexibility.

To deal with the limited resources on motes, TinyOS needed to be lightweight. To ensure that it was, the authors implemented three metrics and ensured they were met: small footprint size of applications, compiler should optimize code, and the overhead should be low. For the first metric, the authors ensured that the base operating system was no more than 400 bytes, with its C runtime primitives being just over 1 kilobyte. This allowed for more space in memory to be used by the actual applications rather than the operating system. As for the second metric, TinyOS works to reduce unused or redundant code using whole-program compilation to get rid of dead code. This allowed up to a 60% size improvement. In terms of minimizing overhead, this is accomplished by virtue of implementing nesC in whole-program analysis. Additionally, TinyOS, by virtue of having only one stack, works towards this purpose as well. By having these attributes, TinyOS is lightweight, which was perfect for motes at the time.

For sensor networks, reactive concurrency was an important step to save resources. This was the preferred operation method due to the single-stack nature of TinyOS. To detail how effective TinyOS's support for concurrency is, four metrics were used: concurrency in applications, support for race detection at compile time, context switching times, and handling current events with real-time constraints. As for the first metric, it was just a matter of determining if the sensor networks in which TinyOS could be applied would exhibit concurrency, or need to react to events as they occur. The example listed is TinyDB, which could benefit from having TinyOS. As for the second metric, the system must support race detection. Race detection determines situations where two events attempt to access the same data and one of the event changes it without coordination to the other event. This can

lead to bugs and unpredictable behavior. TinyOS performs race detection by seeing events that access and can write to the same variable and warning the programmer to protect these variables with atomic sections, since atomic sections temporarily disable interrupts. For the third metric, context switching times must be studied. From there, the researchers detailed the number of clock cycles to handle interrupts and task switching. For the fourth metric, real-time requirements in the sensor network domain had to be taken into consideration. To meet these requirements, TinyOS handles critical operations by interrupt-driven events and allowing longer tasks to break off from the main stack. This allows TinyOS to have low-latency and respond to external stimuli quickly.

Flexibility is another aspect that TinyOS tried to implement. TinyOS achieves this flexibility by allowing applications to be built from many fine-grained components that can be selected within the system. This allows for adaptability depending on the task, since the system can be changed based on the specification of the project.

III. APPLICATION

TinyOS led to many developments in different areas, some of which being habitat monitoring, object tracking, and TinyDB. For some of these, the implementation of TinyOS benefited from the reactive concurrency that TinyOS provided, while others benefited from low-power consumption, allowing for increased scalability in larger projects.

IV. CONCLUSION

The development of TinyOS created an operating system that primarily benefited sensor networks. One of the issues for sensor networks is the limited resources of motes. To deal with this, TinyOS decreased the resources spent by removing unnecessary lines when compiling code, among other things. Another issue for sensor networks is flexibility, which TinyOS was able to deal with due to its modular nature. This allowed TinyOS to adapt to any task it was given. Because of these advantages, TinyOS was able to be implemented in a way that benefited many different areas, some of which being habitat monitoring, object detection, and TinyDB.

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